

Algebra 1
Unit 13
Lesson H2 Practice

Name Answer Key
Date _____
Period _____

Use the functions $b(x) = -2x^2 + 6x - 3$ and $c(x) = 8 + 3x$ to determine each new function. Write your answer in standard form.

$$\begin{aligned} 1. \quad (b + c)(x) &= -2x^2 + 6x - 3 \\ &\quad + 3x + 8 \\ (b + c)(x) &= -2x^2 + 9x + 5 \end{aligned}$$

$$\begin{aligned} 2. \quad (b - c)(x) &= (-2x^2 + 6x - 3) - (8 + 3x) \\ &= -2x^2 + 6x - 3x - 3 - 8 \\ (b - c)(x) &= -2x^2 + 3x - 11 \end{aligned}$$

$$\begin{aligned} 3. \quad (c - b)(x) &= (8 + 3x) - (-2x^2 + 6x - 3) \\ &= 8 + 3x + 2x^2 - 6x + 3 \\ (c - b)(x) &= 2x^2 - 3x + 11 \end{aligned}$$

$$\begin{aligned} 4. \quad (b \cdot c)(x) &= (-2x^2 + 6x - 3)(8 + 3x) \\ &= -16x^2 + 48x - 24 + (-6x^3) + 18x^2 - 9x \\ (b \cdot c)(x) &= -6x^3 + 2x^2 + 39x - 24 \end{aligned}$$

$$\begin{aligned} 5. \quad \left(\frac{b}{c}\right)(x) &= \frac{-2x^2 + 6x - 3}{3x + 8} \end{aligned}$$

6. Write the restriction on the domain for $\left(\frac{b}{c}\right)(x)$.

$$x \neq -\frac{8}{3}$$

$$3x + 8 \neq 0$$

$$3x \neq -8$$

$$x \neq -\frac{8}{3}$$

Paulina and Stefan are renting a boat for a trip to the Keys. The boat costs \$200 per day. The rental includes 125 miles for free and then charges \$0.50 per mile. The function $B(m) = 0.50(m - 125) + 200$ models the cost of the boat, B , in terms of miles traveled, m . They determine gas will cost \$1.75 for each mile they travel. The company charges a \$75 fee for the first tank of gas. The function $F(m) = 1.75m + 75$ models the cost of gasoline for their vacation.

7. What is a reasonable domain for $B(m)$?

Answers may vary.

Must include $x \geq 0$, may include a reasonable maximum as well.

8. What is a reasonable domain for $F(m)$?

Answers may vary.

Must include $x \geq 0$

9. Determine the function that gives the total cost, $C(m)$, in dollars, based on the miles traveled.

$$C(m) = B(m) + F(m)$$

$$= 0.50(m - 125) + 200 + 1.75m + 75$$

$$= 0.50m - 62.5 + 1.75m + 275$$

$$C(m) = 2.25m + 212.50$$

10. What is a reasonable domain for $C(m)$?

Answers may vary.

Must include $x \geq 0$